

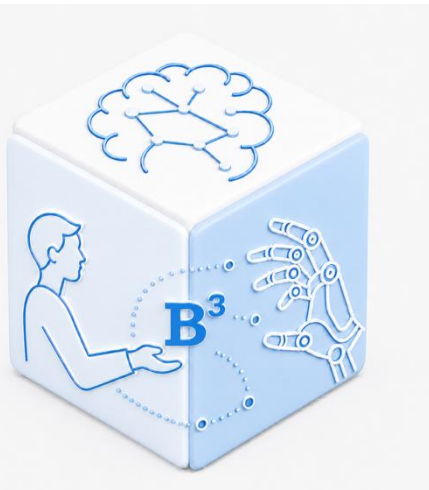
Embodied AI with Web-Scale Video Data

EN.601.439/639

Instructor: Homanga Bharadhwaj

Lecture 2

embodied-ai-with-videos.github.io



Minds



What is a mind?

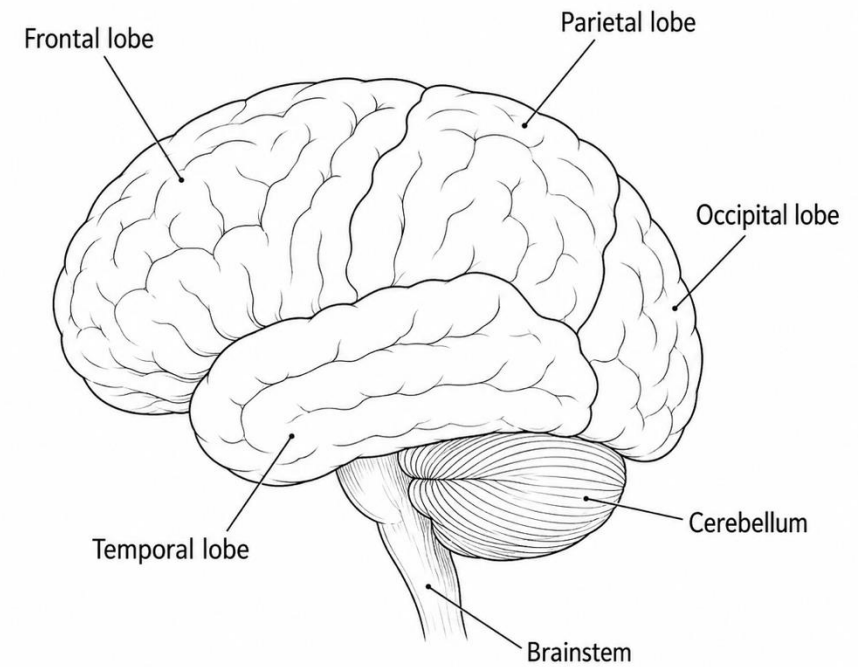
Minds



What is a mind?

A question that has eluded scientists and philosophers alike for generations...

Minds vs. Brains



Core knowledge

A small set of constrained representational systems may provide starting points for flexible learning.

Core knowledge

What is innate vs. what is learned?

Core knowledge

What is innate vs. what is learned?



Babies intuitively understand the world

How do they do it? Given modern objects look very different from objects 10000+ years ago!

Core knowledge

Violation-of-Expectation experiments

Infants are shown an event that either follows physical laws (**expected event**) or violates them (**unexpected event**).

Because infants look longer at things that surprise, puzzle, or contradict their underlying assumptions, a significantly longer looking time at the unexpected event indicates that they possess the conceptual rule being tested.

Core knowledge

Violation-of-Expectation experiments

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Habituation/Dehabituation experiments

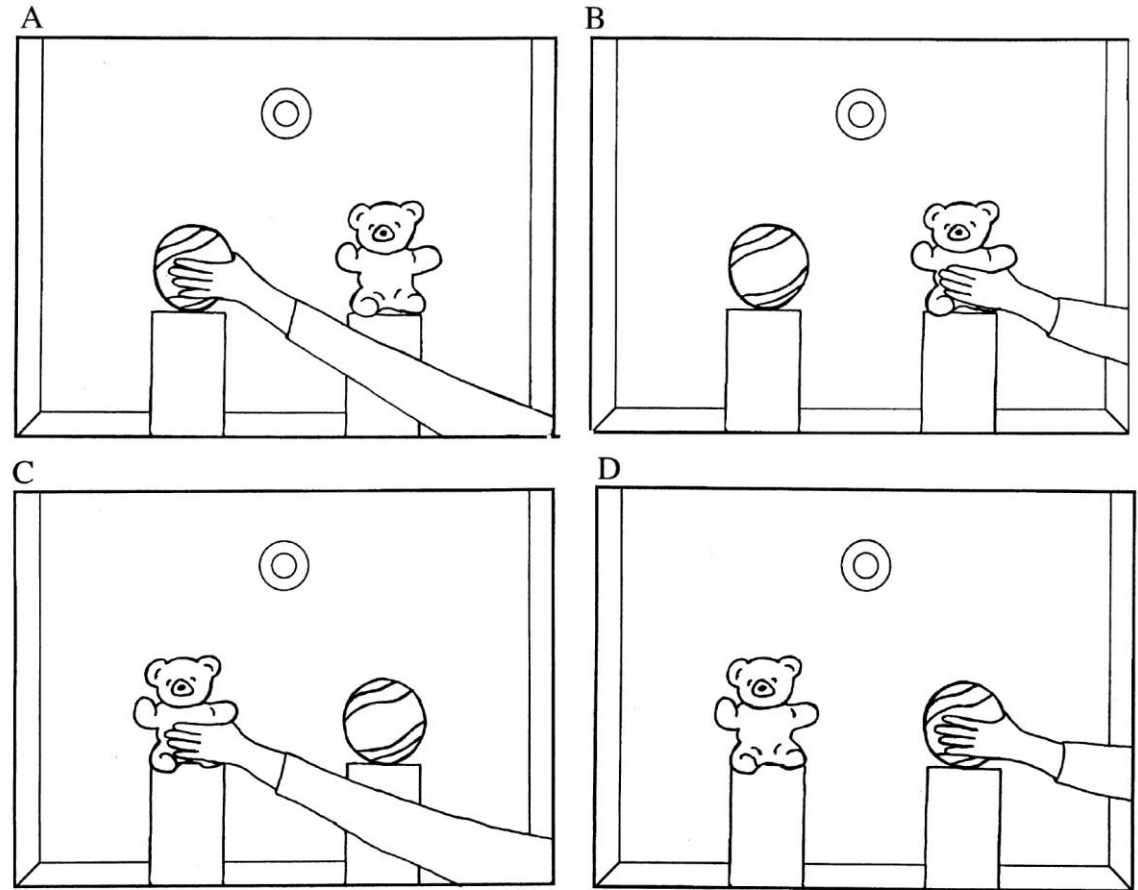
Infants are repeatedly shown a stimulus until their looking time drops (habituation)

A new stimulus is then introduced. If looking time spikes (dehabituation), it indicates the infant perceives a meaningful difference.

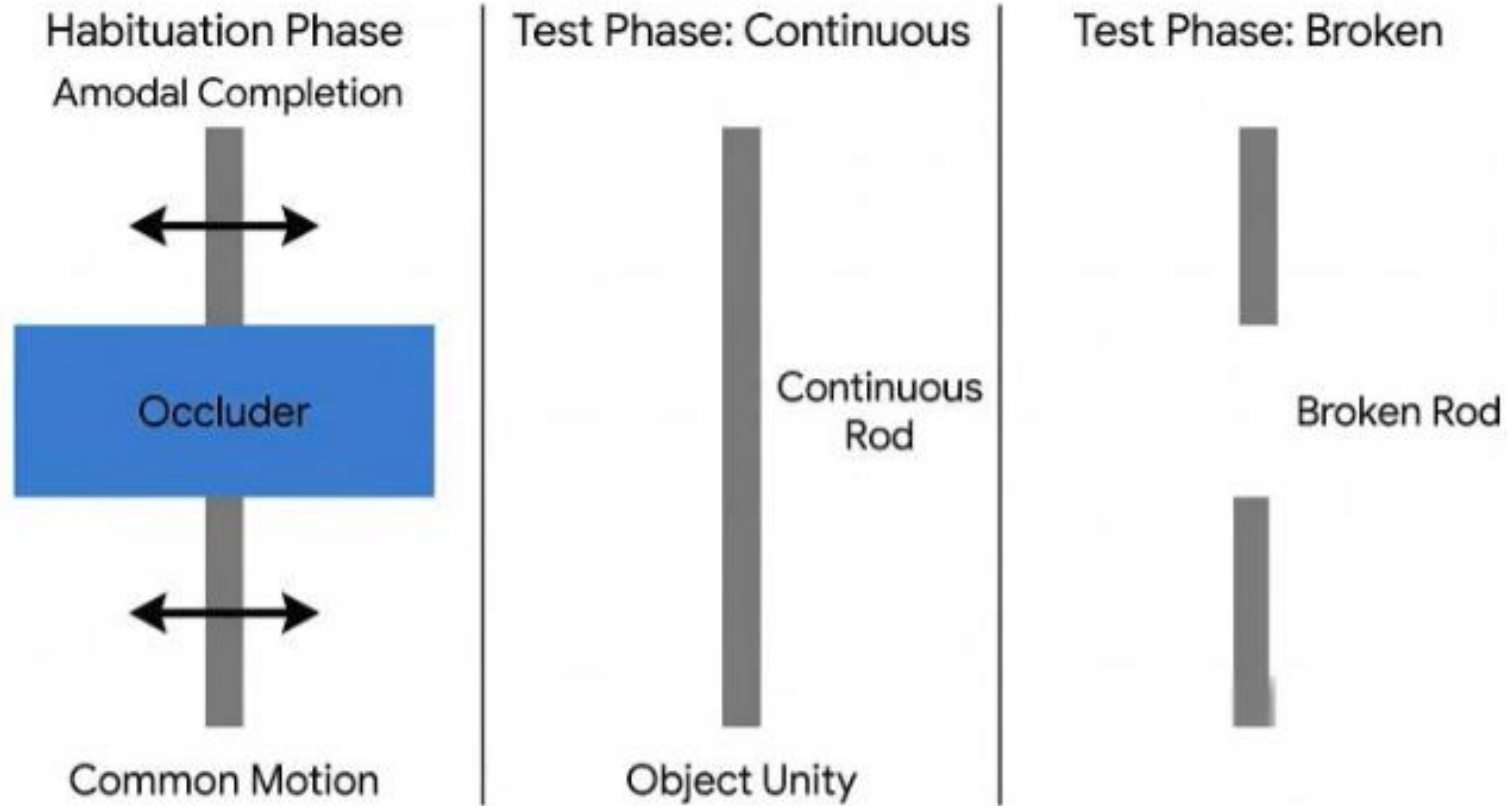
Core knowledge

Infants do not merely track physical paths; they track intentional mental goals.

When a teddy bear and ball switch locations, infants show minimal surprise when the hand takes a new mechanical trajectory to grab the same object, but they look significantly longer if the hand reaches for a brand-new goal.



Core knowledge



Core objects: what counts as one thing?

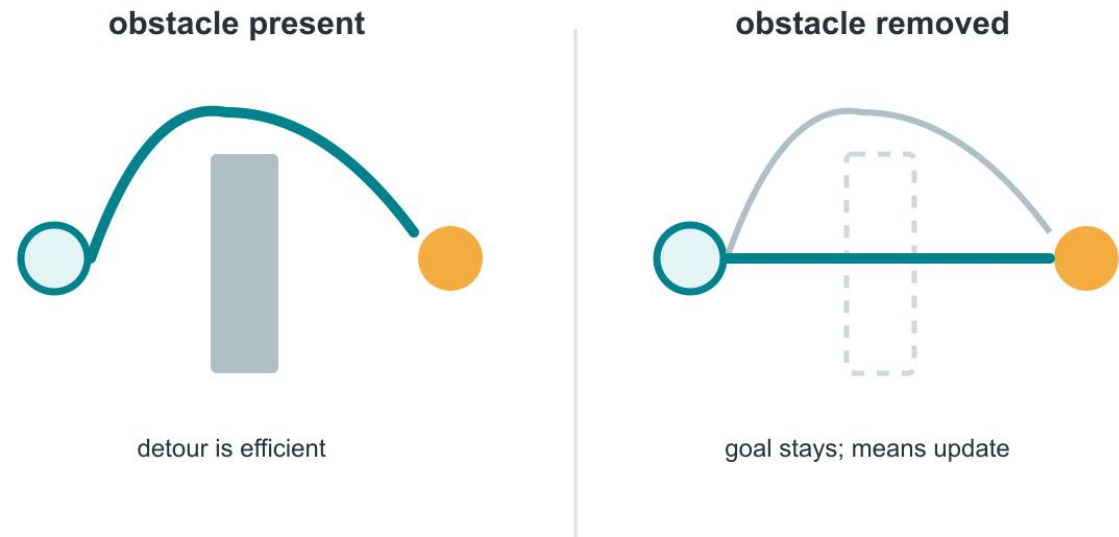
Cohesion → connected, bounded wholes

Continuity → connected paths and persistence behind occlusion

Contact → physical objects do not interact at a distance

Core agents: behavior is organized by goals

- Agents need not obey object cohesion, continuity, or contact.
- Actions are interpreted as goal-directed and means-efficient.
- Contingency, reciprocity, and gaze are inferable cues.
- Obstacles change the expected path, not the inferred goal.



Core number: approximate and ratio-limited

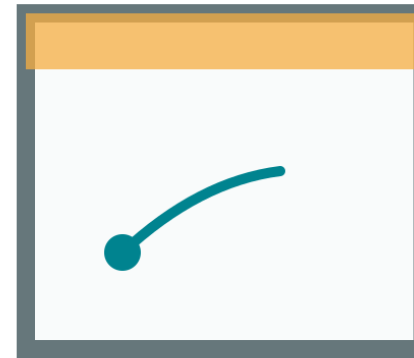
- Abstract across objects, sounds, and actions.
- Discrimination improves from roughly 2:1 at 6 months to 1.15–1.3:1 in adults.
- Supports approximate comparison, addition, and subtraction.



Discriminability follows the ratio between quantities—not their absolute difference.

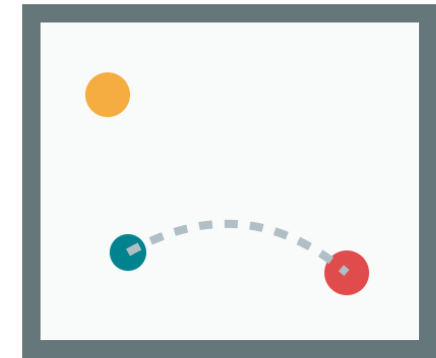
Core space: geometry for reorientation

- Represents distance, angle, and left/right sense among extended surfaces.
- Surface geometry dominates after disorientation.
- Color, odor, and movable landmarks are handled separately.
- Similar signatures appear across ages, species, and cultures.



surface geometry

distance · angle · left/right sense



movable landmarks

handled by a separate association process

Core knowledge is narrow

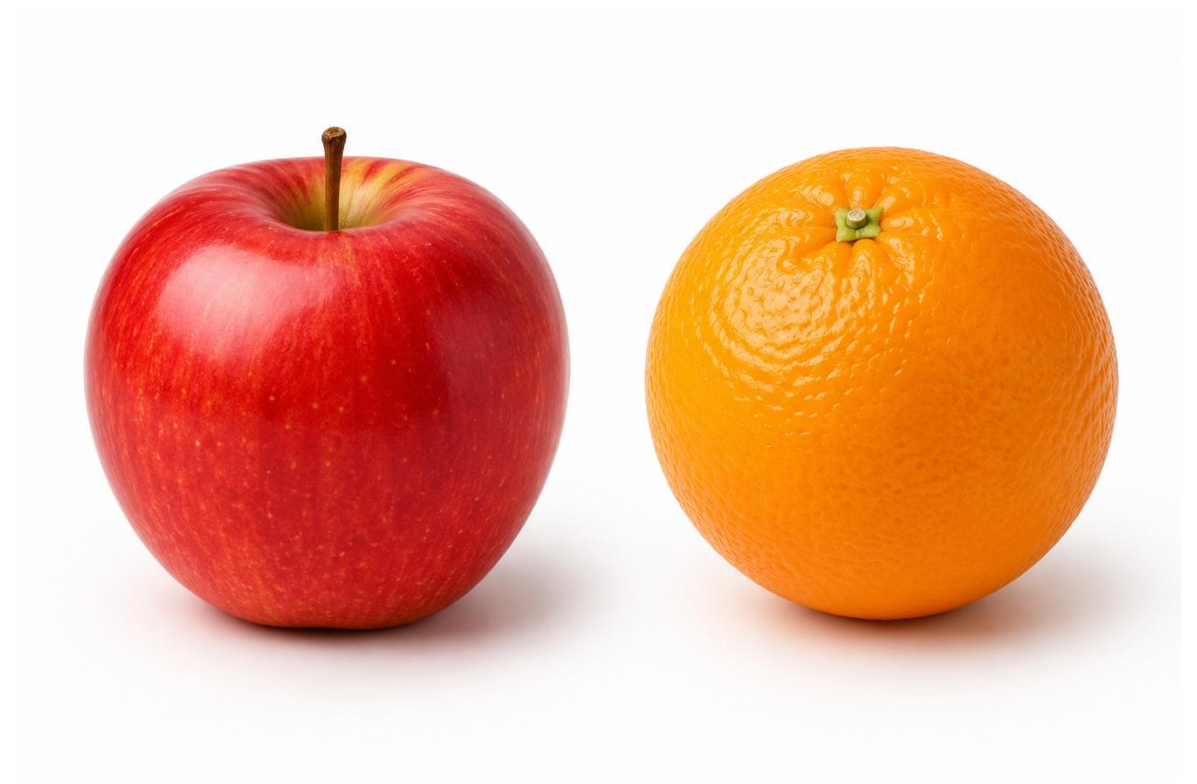
**Core knowledge arguably remains
invariant over generations**

Objects, agents, number, space, forms, and social partners isolate different entities and relations.

Possible explanation for why babies can so efficiently understand about *modern* objects that didn't exist even in the recent past

Concepts

Apple vs. Orange



Concepts

Apple vs. Orange



Fruits vs. Vegetables

Tables vs. AirPods



The representation of concepts

Prototype Theory

Each category is represented by a **prototype**: a summary of its typical features.

Features are **weighted by how often they occur** in the category (Smith & Medin, 1981).

A new example is classified by finding the **most similar prototype**.



Example: To identify this animal, we compare it with stored category prototypes. Its closest match is **bird**.

The representation of concepts

Exemplar theory: store instances, not averages

REPRESENTATION

Store multiple episodes with perceptual, contextual, and action features—no single average is required.

INFERENCE

Retrieve similar exemplars using task-dependent attention.
The same object can support different categories as the relevant features change.

TRADE-OFF

Captures exceptions, but relies on memory, retrieval, and a learned similarity metric.

The representation of concepts

Retrieval vs. Instinct

Some would immediately recognize this bird and others won't



Muscle memory Requires minimal retrieval



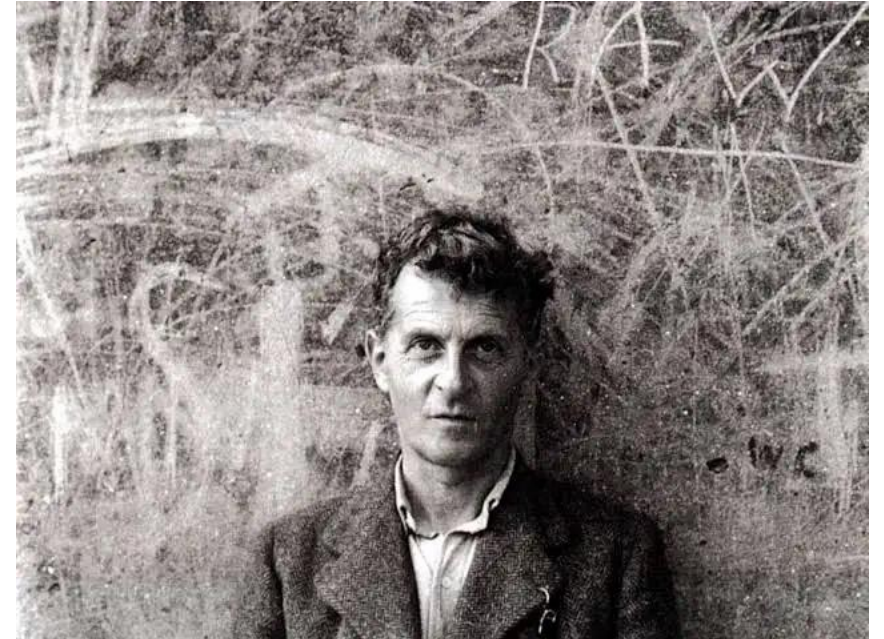
Instincts are contextual on person / environment /...

What is embodied about cognition?

If I give someone the order “fetch me a red flower from that meadow”, how is he to know what sort of flower to bring, as I have only given him a *word*?

Now the answer one might suggest first is that he went to look for a red flower carrying a red image in his mind, and comparing it with the flowers to see which of them had the color of the image. ... But this is not the only way of searching and it isn't the usual way. We go, look about us, walk up to a flower and pick it, without comparing it to anything. To see that the process of obeying the order can be of this kind, consider the order “*imagine* a red patch.” You are not tempted in this case to think that *before* obeying you must have imagined a red patch to serve as a pattern for the red patch which you were ordered to imagine.

(Wittgenstein, 1958)



What is embodied about cognition?

We know that the conceptual system engages, online, with sensory and motor content

How is it that the human conceptual system is able to disengage from the sensorimotor system?

Answering this question would say something about how the human mind is able to *detach from the present and extrapolate from finite experience to hypothetical states* of how the world could be.

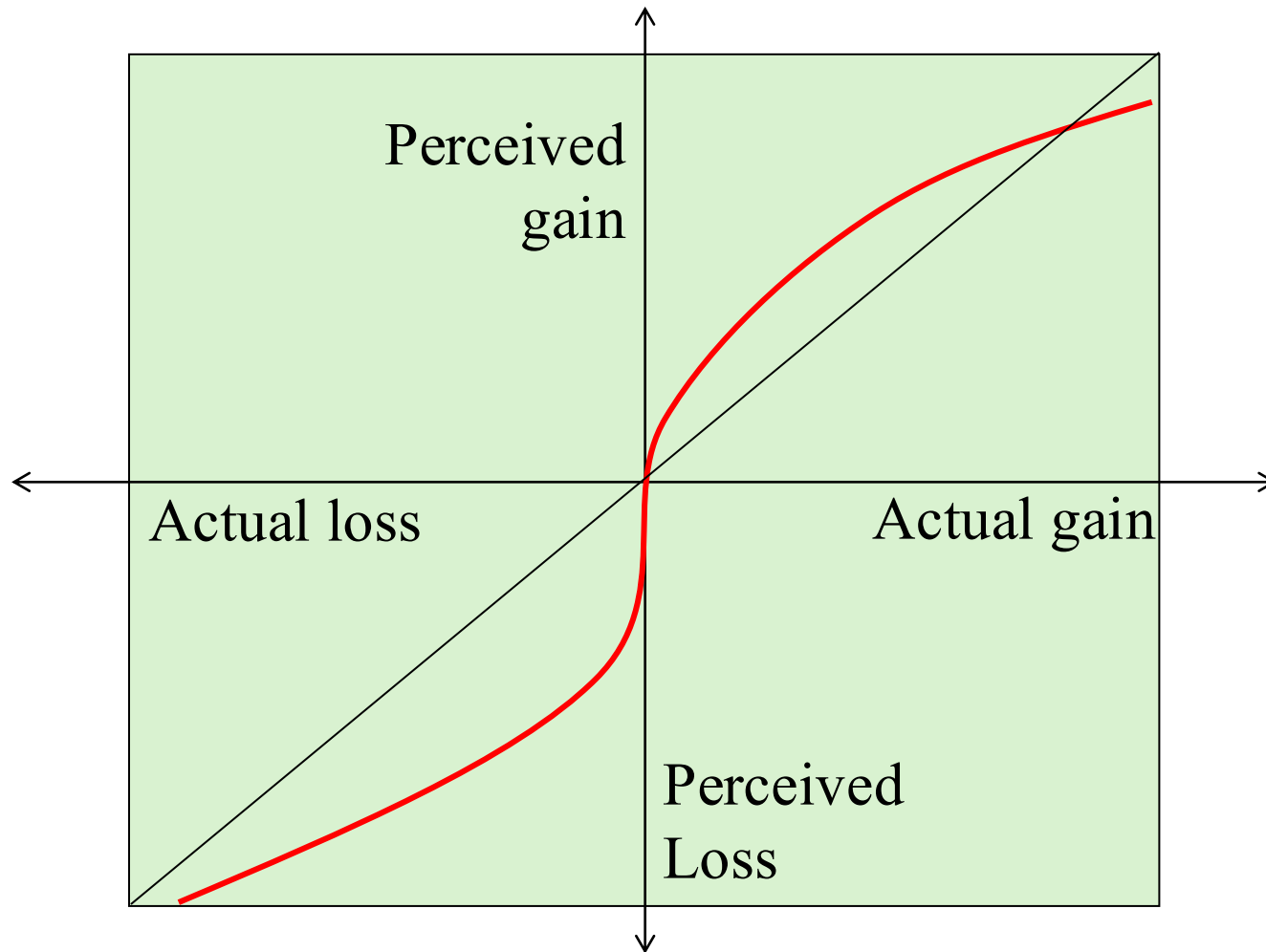
What is embodied about cognition?

It is the ***independence of thought from perception and action*** that makes human cognition special

and that independence is guaranteed by the representational distinction between concepts and sensorimotor representations.

Heuristics in Decision-making

Prospect theory

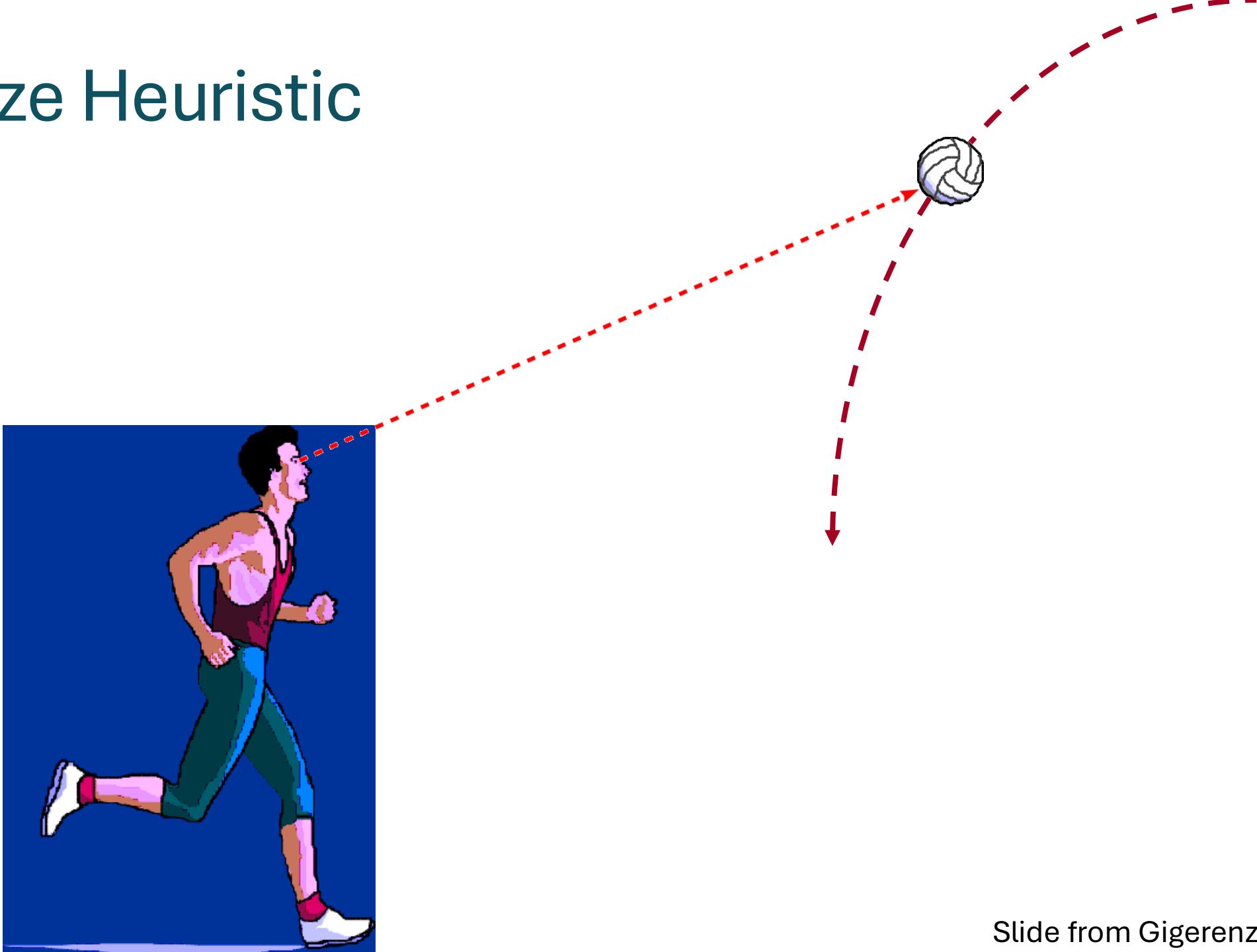


- Initial gains feel especially valuable
- Sensitivity diminishes as gains increase
- *Losses have a strong immediate impact*
- Sensitivity diminishes as losses increase

Typical pattern

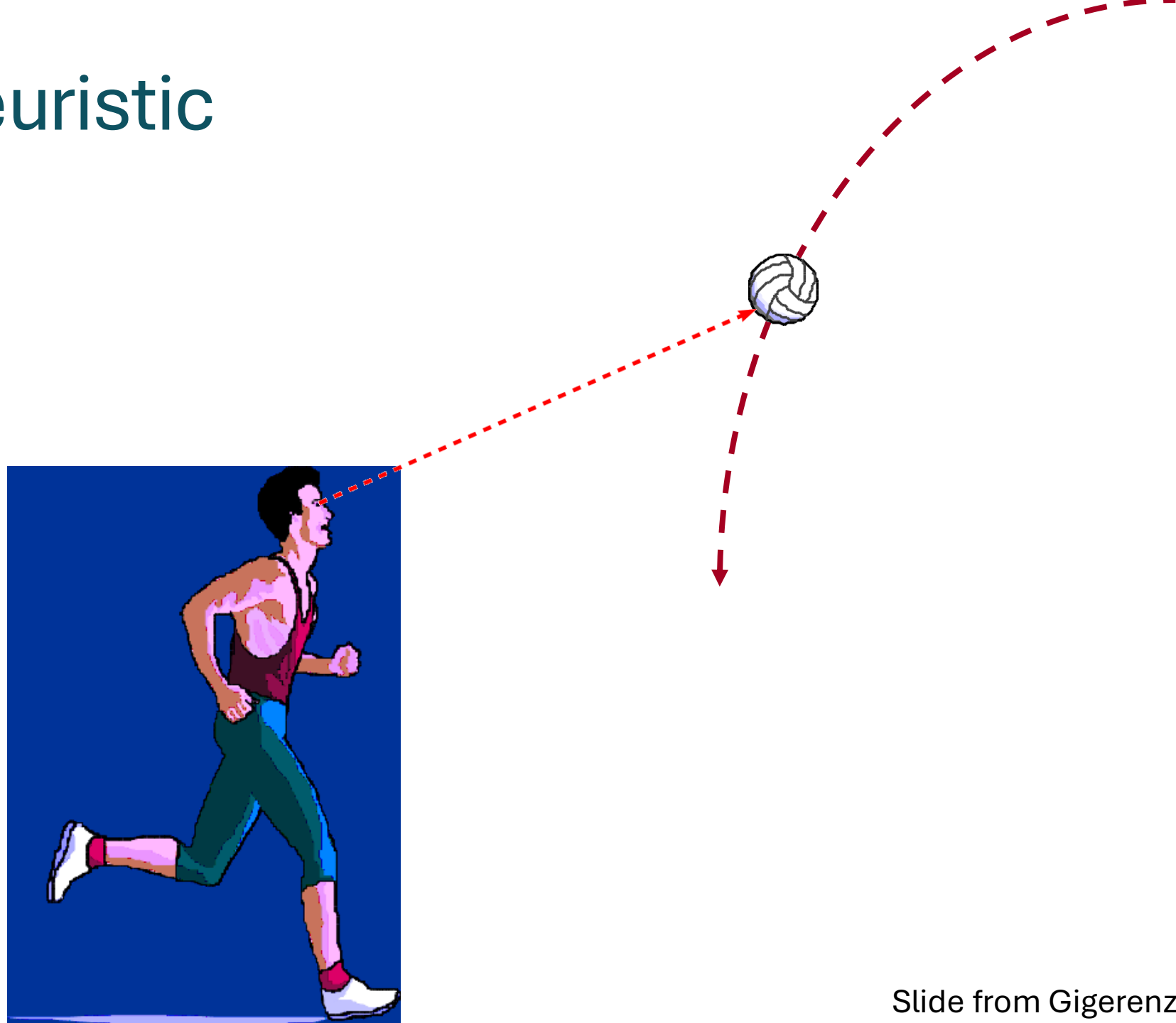
- **Loss aversion:** losses loom larger than equivalent gains
- **For gains:** often risk averse
- **For losses:** often risk seeking

Gaze Heuristic



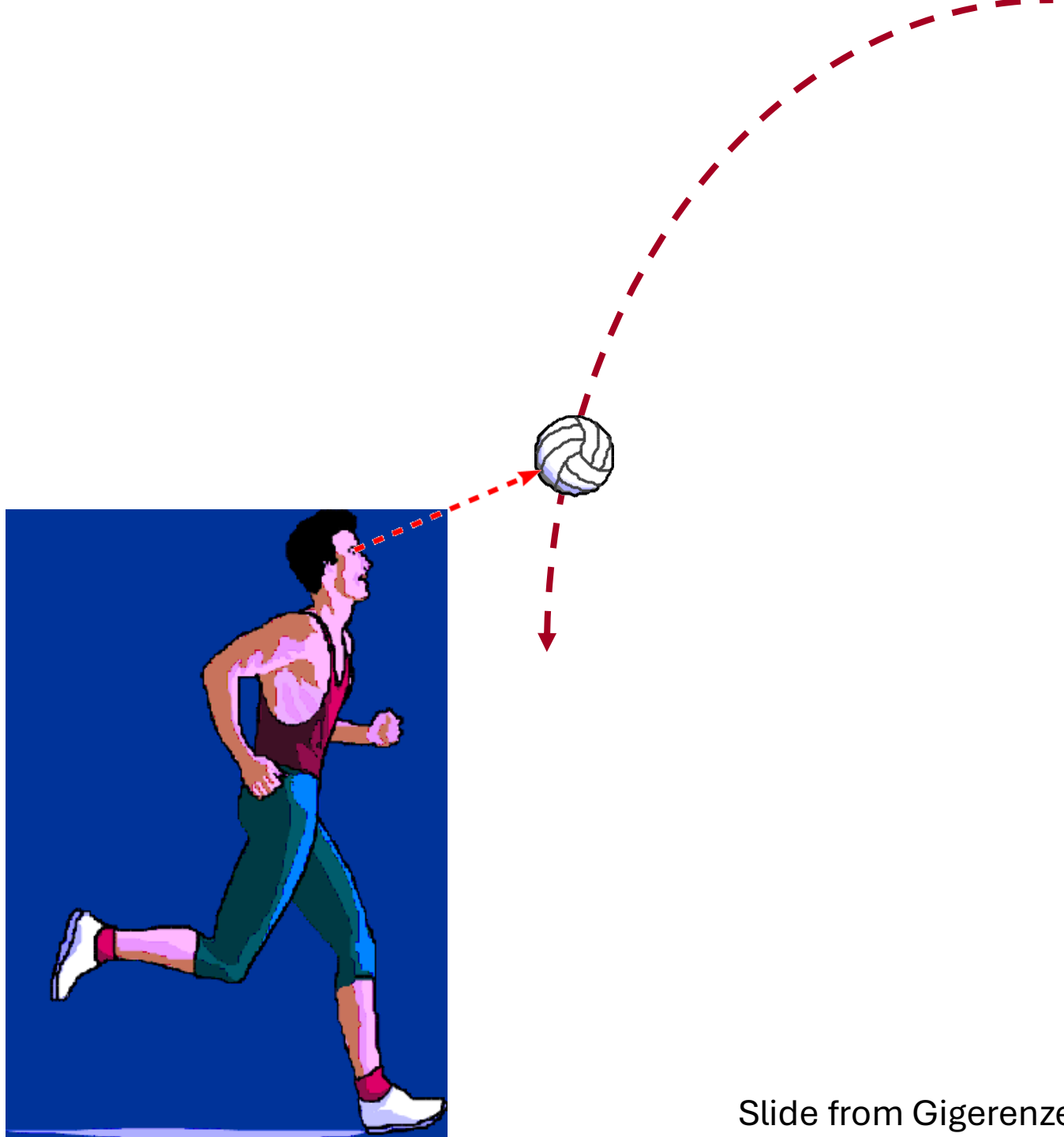
Slide from Gigerenzer

Gaze Heuristic



Slide from Gigerenzer

Gaze Heuristic



Slide from Gigerenzer

Gaze Heuristic



Slide from Gigerenzer

Gaze Heuristic

Where do these heuristics come from?

- *Predation and pursuit:*
bats, birds, dragonflies, hoverflies, teleost fish, houseflies
- *Avoiding collisions:*
sailors, aircraft pilots
- *Sports:*
baseball outfielders, cricket, dogs catching Frisbees

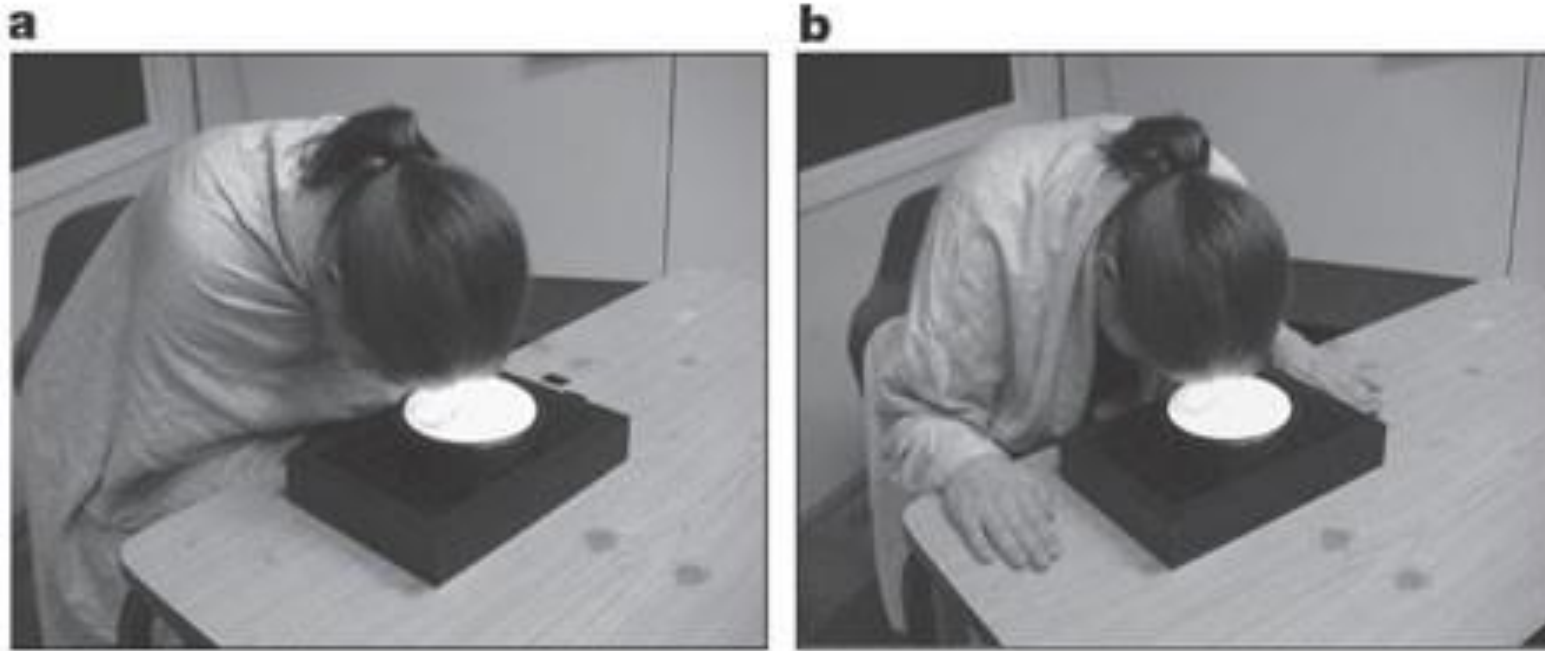


Gaze heuristic ignores all causal relevant variables

Rational Imitation

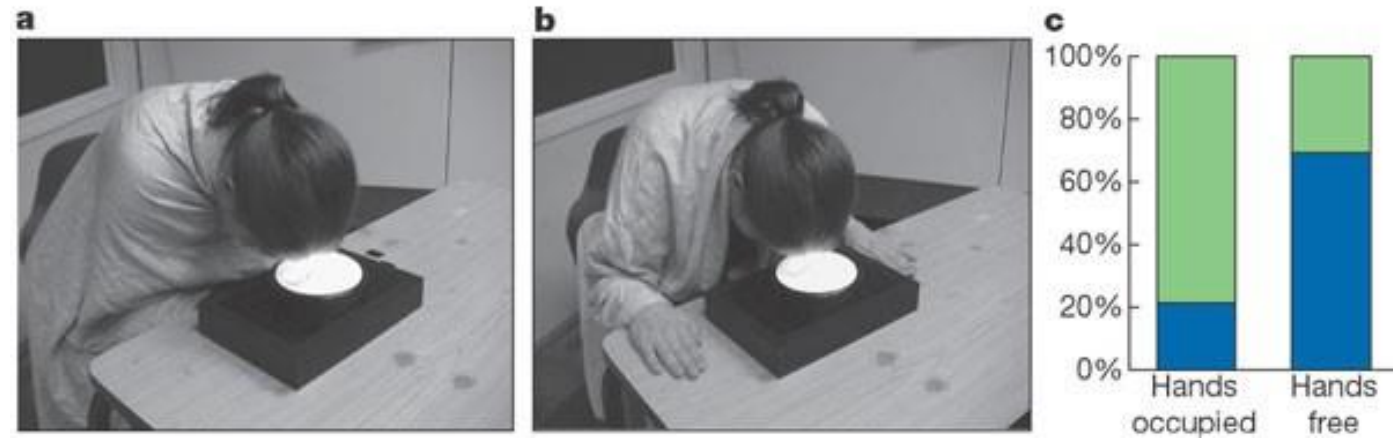
A demonstration is not a trajectory to mimic; it is evidence about goals, constraints, and efficient action.

How do babies imitate?



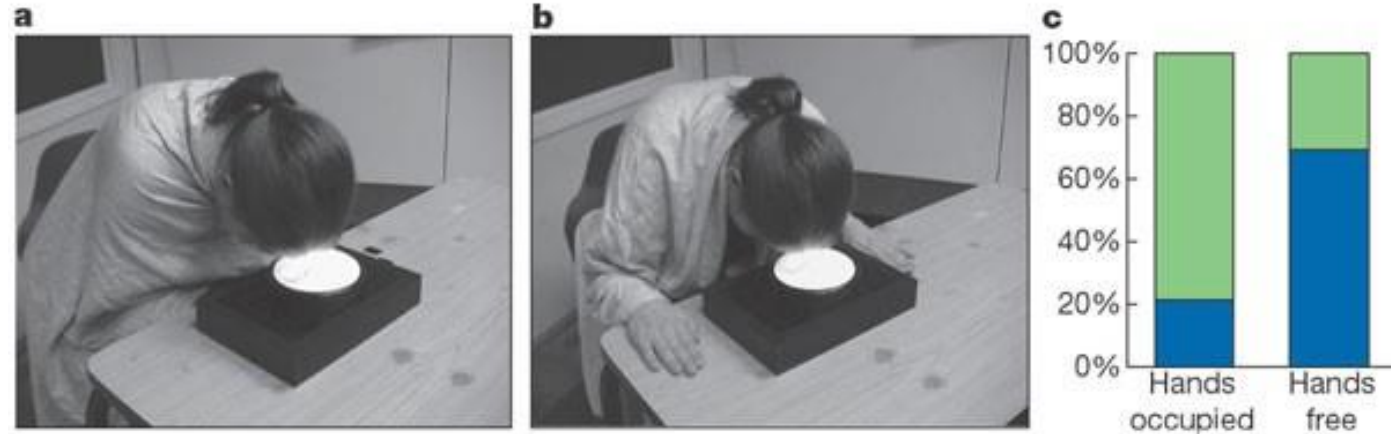
One unusual demonstration, two contexts

- Fourteen-month-olds watched an adult activate a light box with her forehead.
- Hands occupied: $n = 14$. Hands free: $n = 13$.
- Test occurred one week later; infants had 20 seconds and their own hands were free.



Context and not the observed motion changes imitation

- Hands free: 69% re-enacted the head action.
- Hands occupied: 21% re-enacted it; $p < .02$.
- All infants also used the efficient hand action.



The unusual means was copied selectively when it looked freely chosen.

Two accounts make opposite predictions

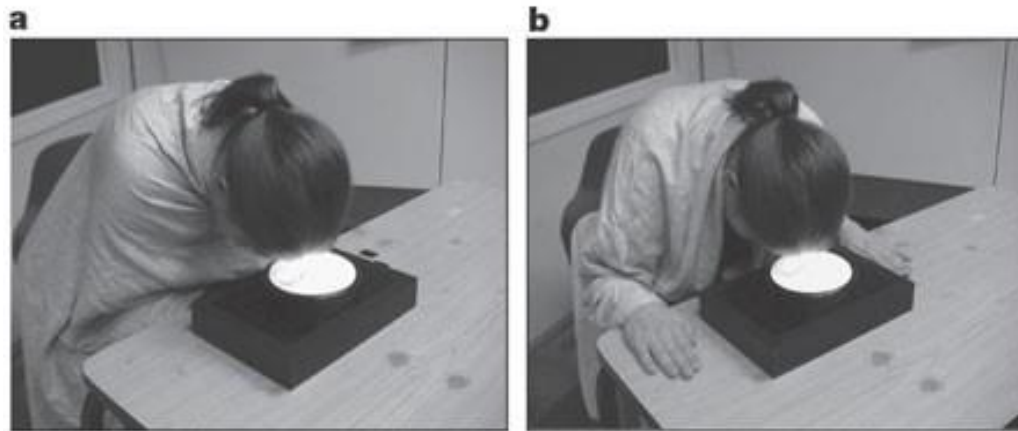
AUTOMATIC MEANS COPYING

The same observed head movement should trigger the same head movement at test regardless of context.

RATIONAL-ACTION INFERENCE

Copy the head action only if the adult appeared to choose it freely. If her hands were occupied, infer a situational constraint and use your own free hands.

Same movement; different inferred reason.



Imitation and emulation coexist

All infants used the efficient hand action at test.

EMULATION

Remember the outcome, illumination by touch, and recruit the most available means.

SELECTIVE IMITATION

Re-enact the unusual head action only when it looked like a rational, freely chosen means.

The head action carries the context-sensitive signal.

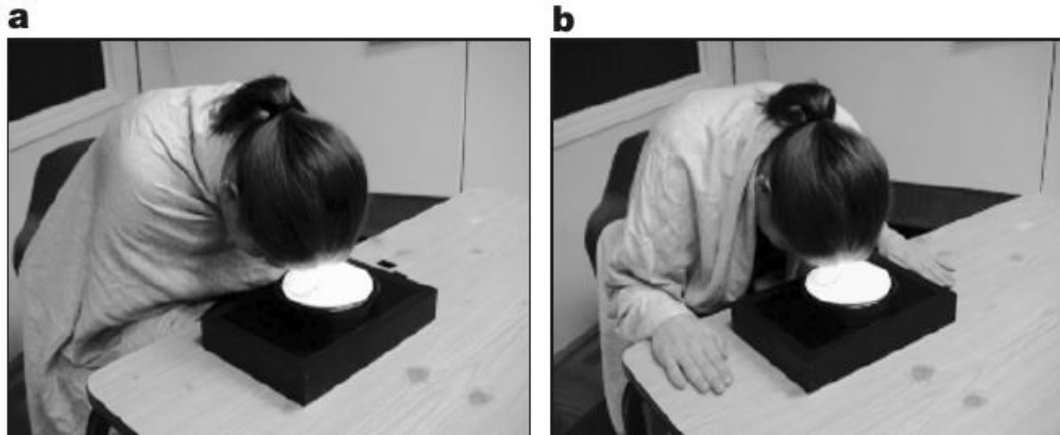
A computational lens of visual imitation

Observe $d = (\text{action}, \text{context})$

Infer $p(\text{goal}, \text{constraint} \mid d) \propto p(\text{action} \mid \text{goal}, \text{constraint}) p(\text{goal}, \text{constraint})$

Act $a^* = \arg \min_a \text{Cost}(a \mid \text{goal}, \text{own constraints})$

Instruction following vs. Curiosity



Learning from instruction

A demonstration provides a trajectory. The learner still has to infer the goal, the relevant means, and the demonstrator's constraints.



Learning from curiosity

A self-chosen action selects the next observation. Informative actions reveal controllable structure and reshape future data.

Strong learners combine guidance with active evidence seeking.

Intrinsic motivation

- Seek reducible prediction error, not random novelty.
- Prefer actions with high expected information gain.
- Reward learning progress: improvement, not surprise alone.
- Every action changes the learner's future data distribution.



Visual Affordances

WHAT DO YOU SEE?



Visual Affordances

- A handle affords grasping.
- A chair affords sitting.
- A gap affords passage only if it is wide enough relative to the agent's body.
- An object's affordances change with the agent's size, capabilities, pose, and goal.



Functional Correspondences



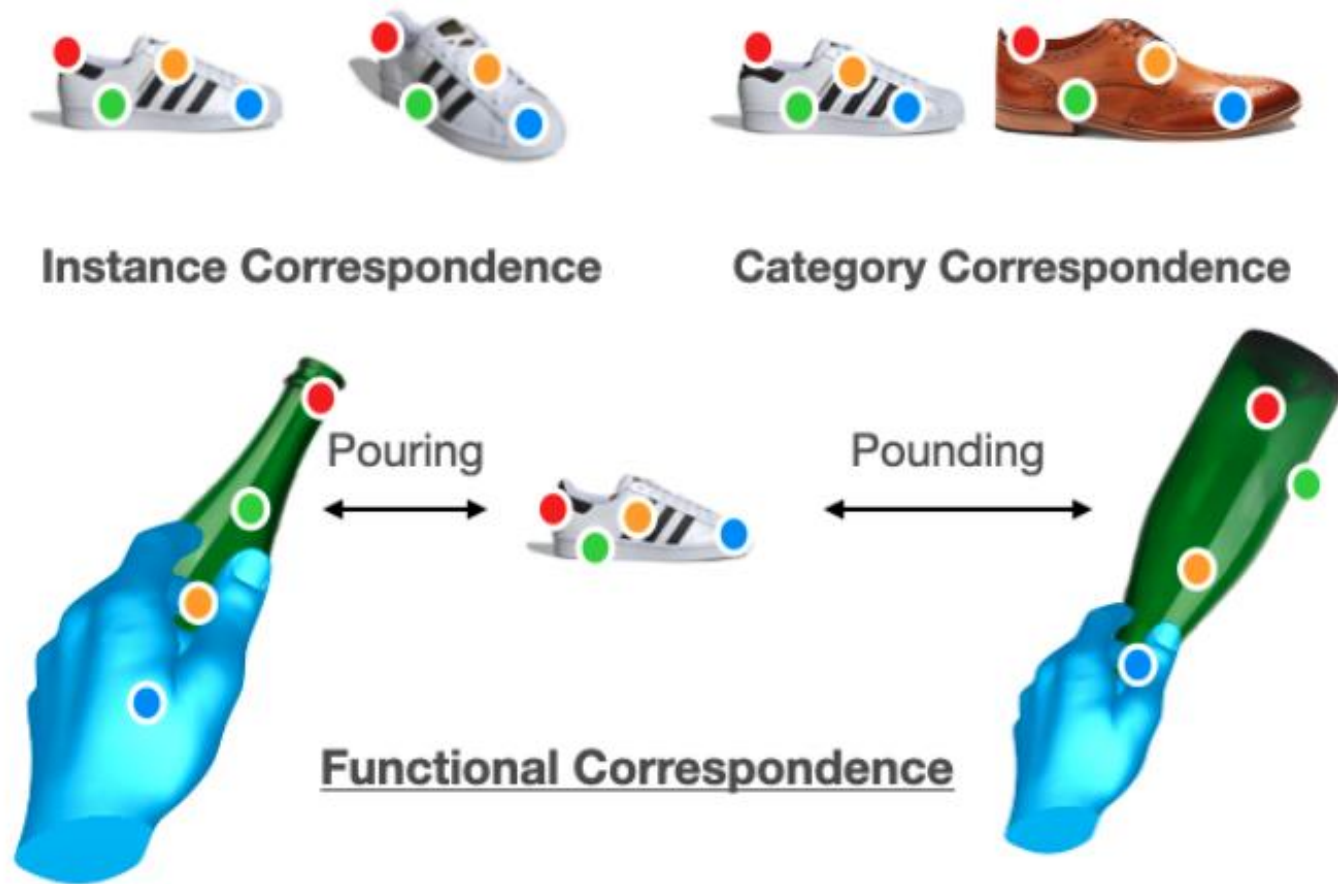
**DO THEY HAVE ANY
COMMON
AFFORDANCE?**

Functional Correspondences



**DO THEY HAVE ANY
COMMON
AFFORDANCE?**

Functional Correspondences



Functional Grasping

Humans grasp objects with a purpose!



Perception-Action Binding

Participants were shown photographs of common household objects with prominent. **The handles pointed either to the left or to the right**

The Task: Participants had to decide whether the object was **upright or inverted** by pressing a key with either their left or right hand.

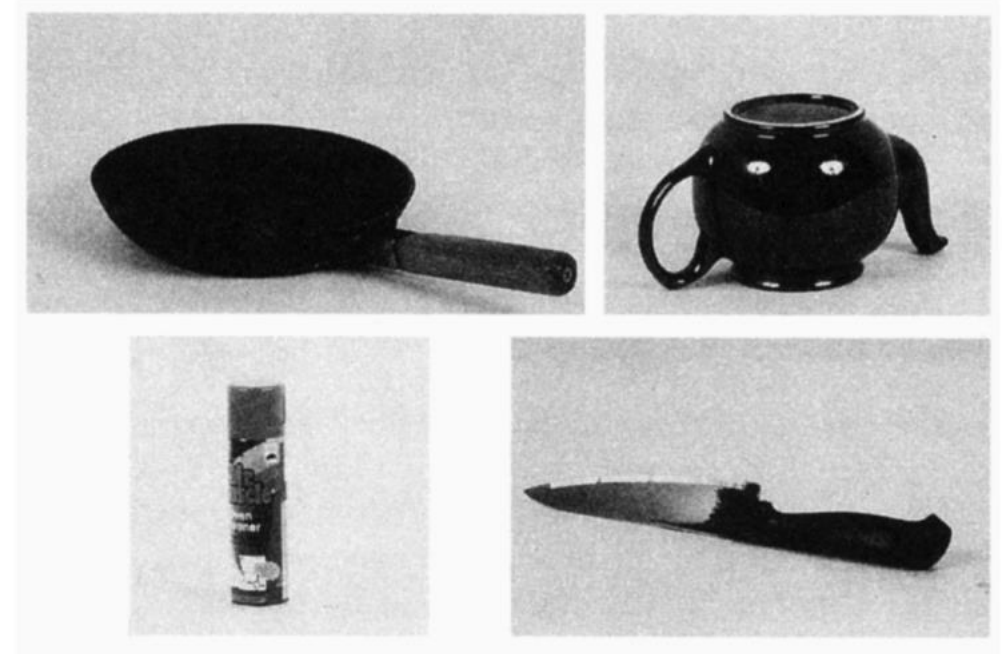


Figure 1. Examples of the stimuli used in the experiments. Experiments 1 and 2: right orientation, upright (frying pan); left orientation, inverted (teapot). Experiment 3: anticlockwise wrist rotation compatibility, inverted (knife); clockwise wrist rotation compatibility, upright (aerosol can).

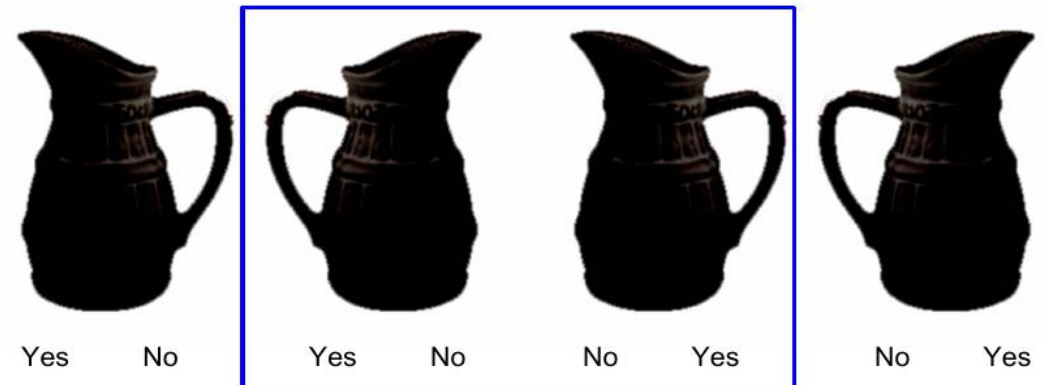
Perception-Action Binding

The Finding: Responses were significantly faster and more accurate when the hand required to make the judgment matched the side of the object's handle

(e.g., pressing the "upright" button with the right hand when looking at a teapot with its handle on the right)

Tucker and Ellis task

Is this a pitcher?



Faster response, when Yes button is on the same side as the handle.

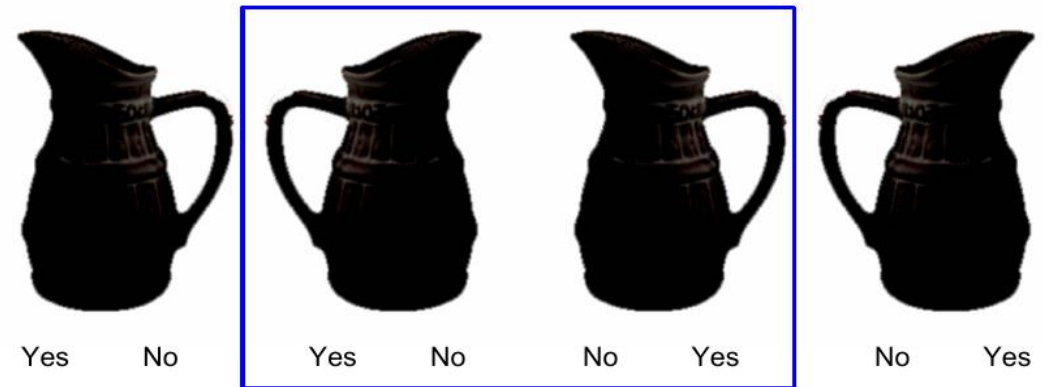
Perception-Action Binding

This phenomenon is known as the **Handle Compatibility Effect**.

It demonstrates that visual perception and motor execution are implicitly bound; seeing the handle automatically "primes" or potentiates the grasping action of the corresponding hand.

Tucker and Ellis task

Is this a pitcher?



Faster response, when Yes button is on the same side as the handle.

Predictive brains, situated agents

How we simulate an apparent reality

Generative models invert hidden causes

Latent causes z generate sensory data y through $p(y | z)$.

Perception approximates the posterior:

$$p(z | y) \propto p(y | z) p(z)$$

SAME SIGNAL → DIFFERENT UNDERLYING CAUSAL MECHANISM



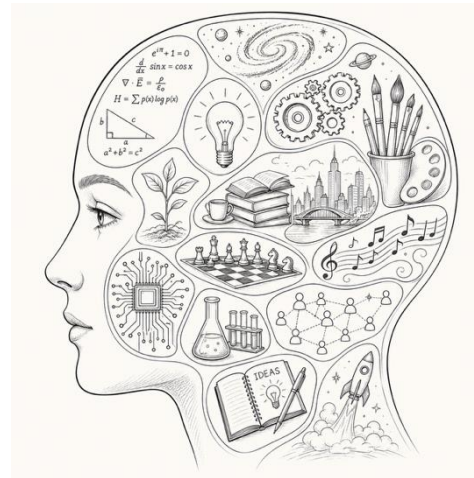
Active inference closes the perception–action loop

Perception

Change internal states to reduce prediction error

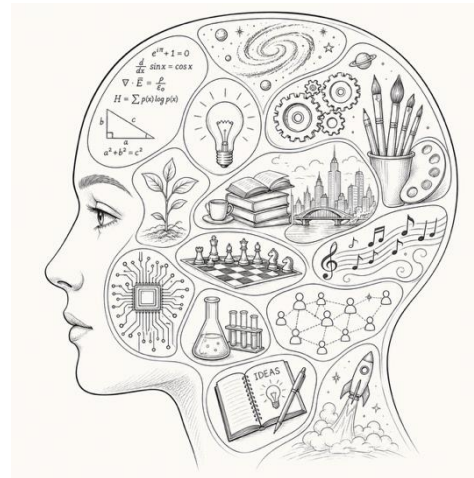
Action

Change sampled sensations to fulfill predictions



predictive processing provides a unifying framework for cognitive science through the minimization of prediction error

Predictive processing in the brain

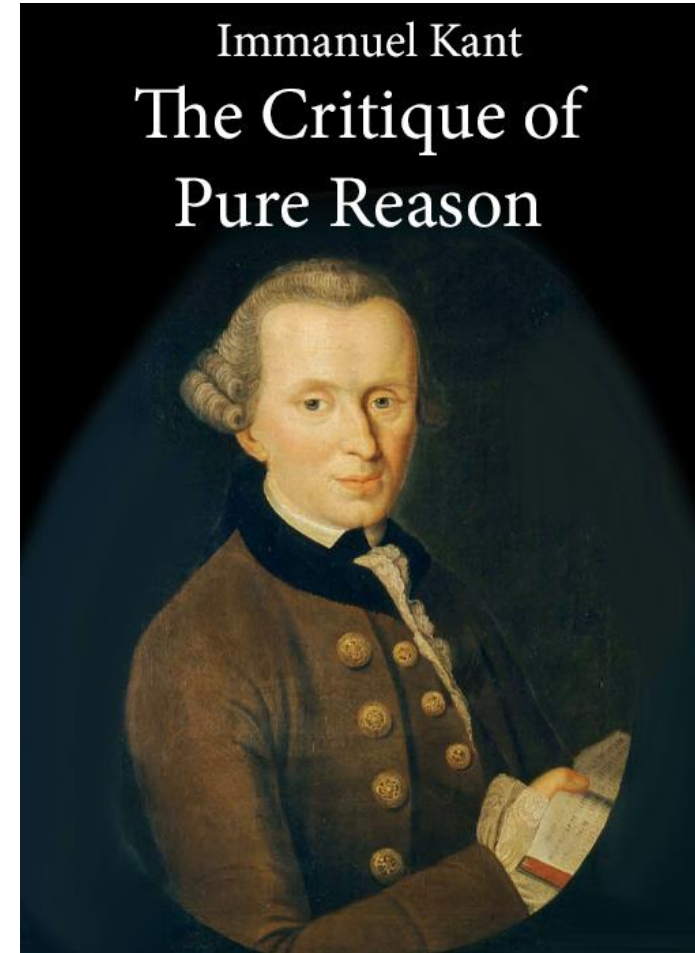


Andy Clark's framework of hierarchical predictive processing models the **brain as an active prediction machine** that minimizes discrepancies between top-down expectations and sensory data.

The mind as a generative engine



Andy Clark



Clark and Kant are centuries apart but share plenty similar ideas

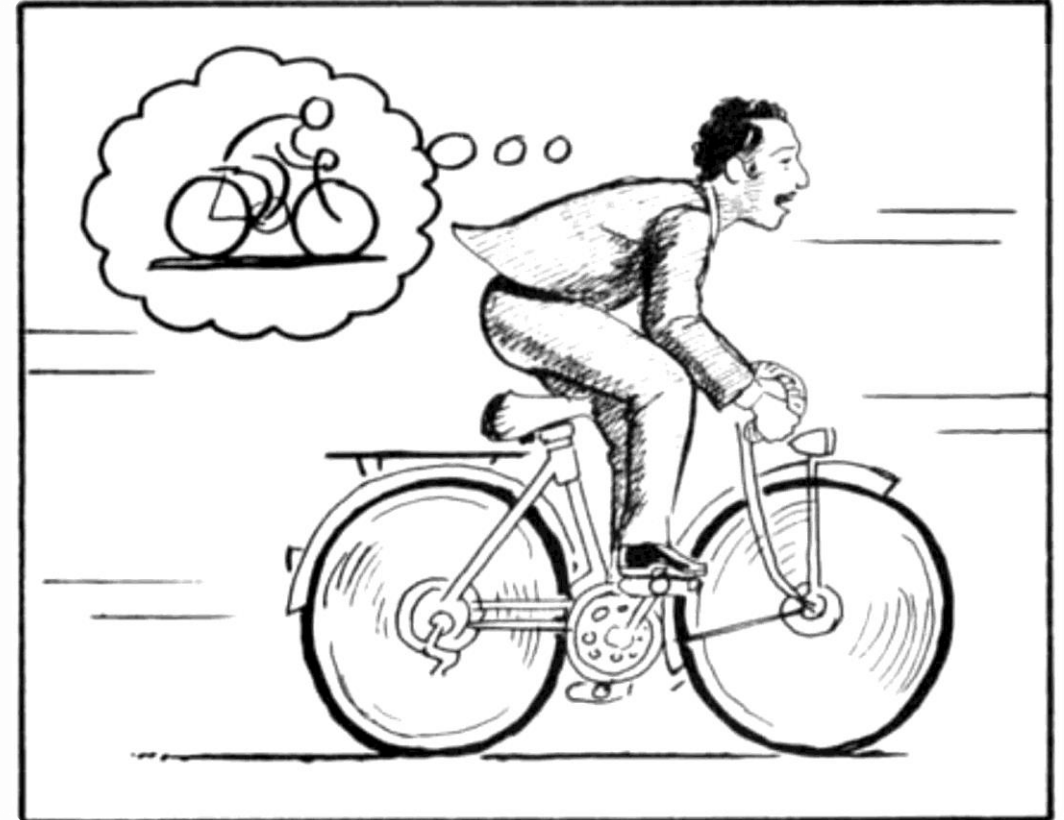
The mind as a generative engine

Kant's View (The Copernican Revolution in Philosophy): Kant argued that we never experience the world exactly as it is in itself (*noumena*). Instead, our minds possess *a priori* (built-in) frameworks like space, time, and causality that actively structure and shape everything we perceive (*phenomena*). Without these mental templates, sensory data would be nothing but "blind" chaos.

Clark's View (The Predictive Brain): Clark translates Kant's philosophical framework into modern neuroscience. He argues that the brain is a **prediction machine**. It doesn't wait for sensory data to arrive; it constantly runs an internal "generative model" that projects top-down predictions onto the world. What we actually experience is what Clark calls a **"controlled hallucination"** i.e. a mix of what the brain expects to happen and a correction mechanism called *prediction error*.

Mental Models

A model of how the world
as we perceive, behaves



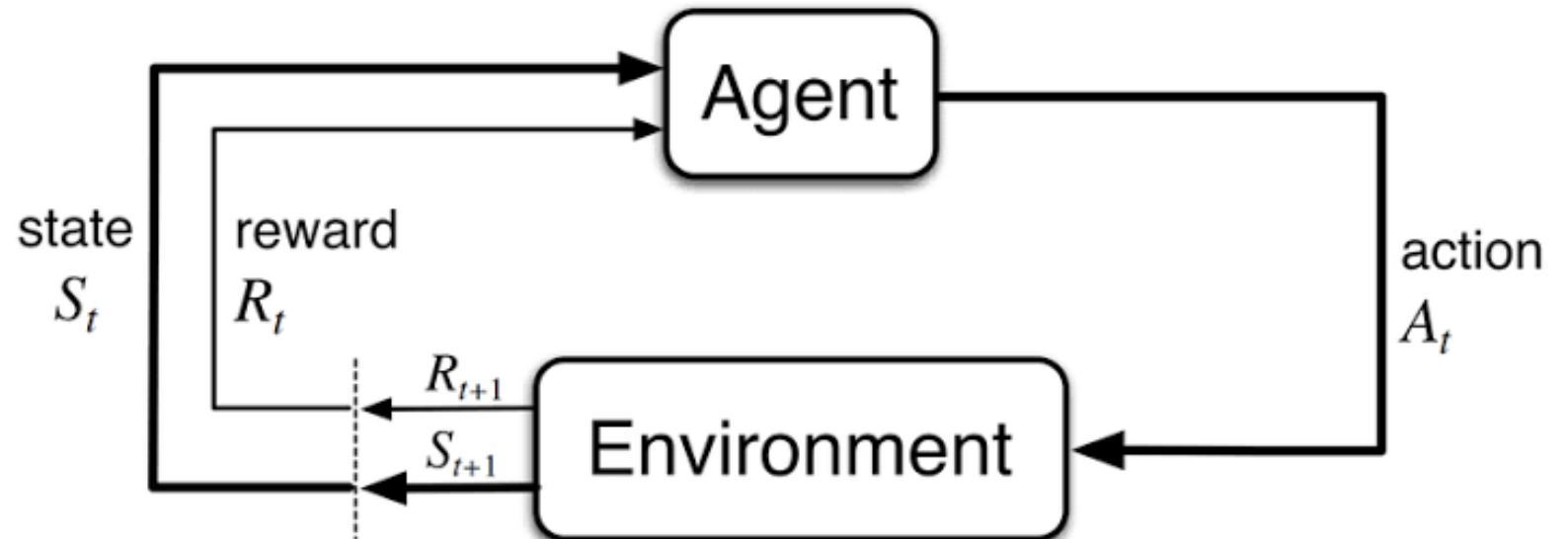
Mental Models

How Do we Build Mental Models?

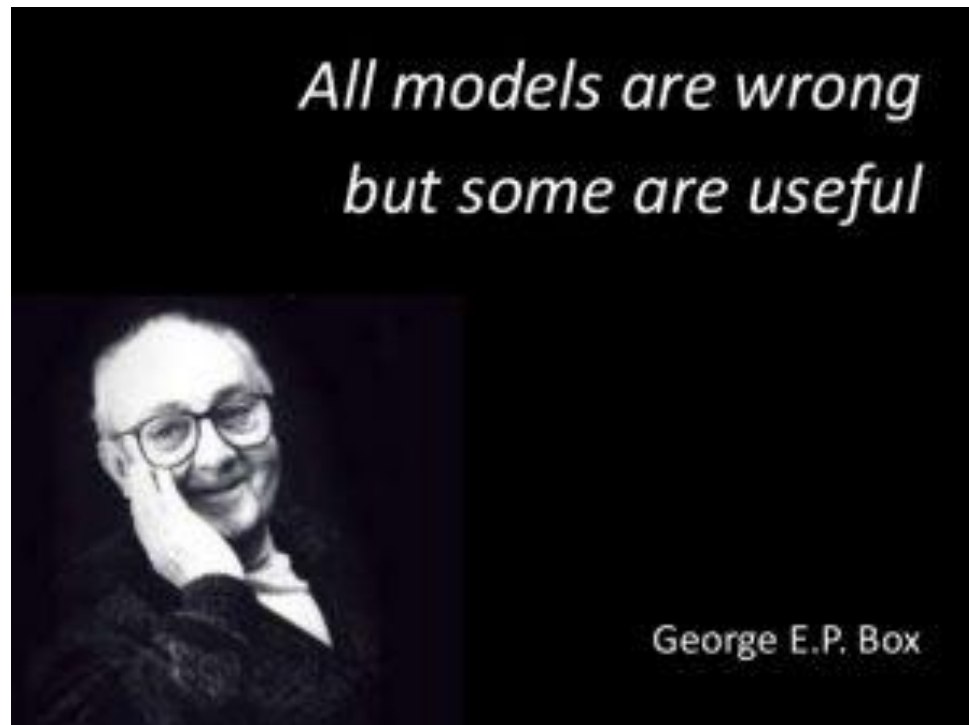
Through:

1) Observation

2) Interaction



Mental Models



So,

*How do we operate almost
perfectly with imperfect
mental models?*

Mental Models



So,

How do we operate almost perfectly with imperfect mental models?

To pour milk from a jar we do not need to perfectly model how the milk will spill!

Intelligence without representation

Brooks's alternative: build complete agents from asynchronous behavior layers coupled continuously to the world.

“The World is its Own Best Model” - Brooks

The World as its Own Model: Instead of building complex abstract maps or internal representations, an agent should use the real world directly as its primary reference point.

“The World is its Own Best Model” - Brooks

Subsumption Architecture: Systems are built using simple, layered behavioral modules (like avoiding obstacles or moving forward) that connect perception directly to action.

“The World is its Own Best Model” - Brooks

Emergent Intelligence: Intelligent behavior emerges dynamically from the parallel interaction of these simple reactive layers with the physical environment, bypassing central symbolic control.

Inside a Brooks abstraction

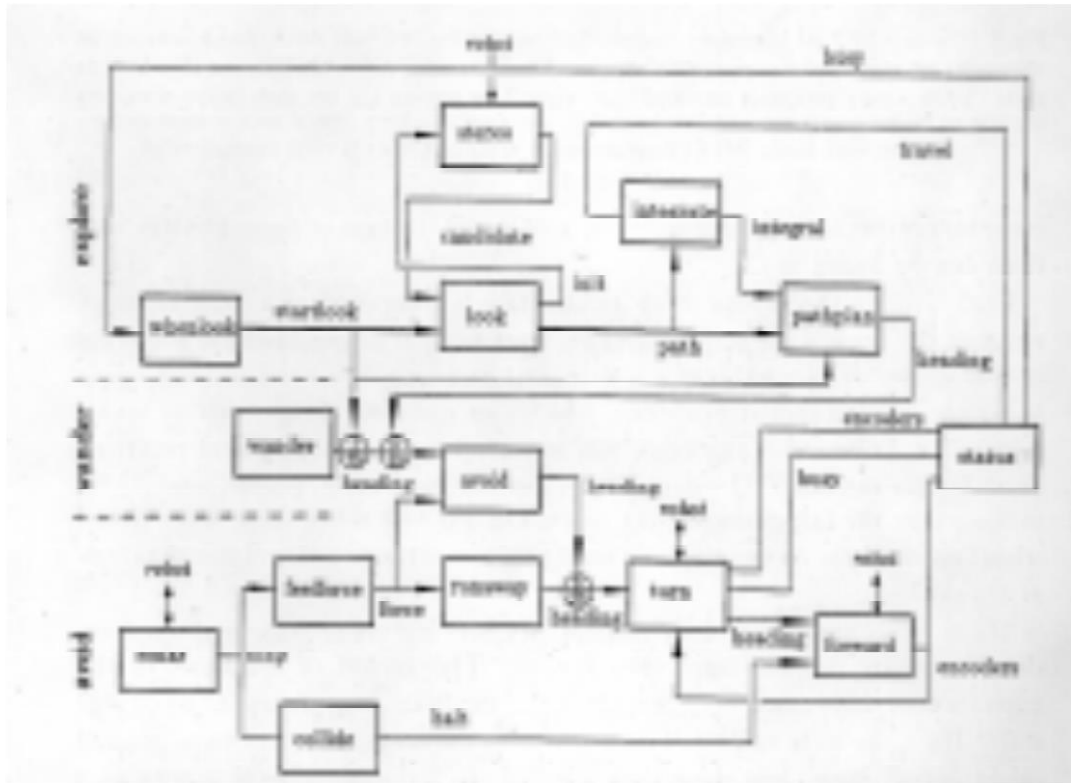


Fig. 2. We wire, finite state machines together into layers of control. Each layer is built on top of existing layers. Lower level layers never rely on the existence of higher level layers.

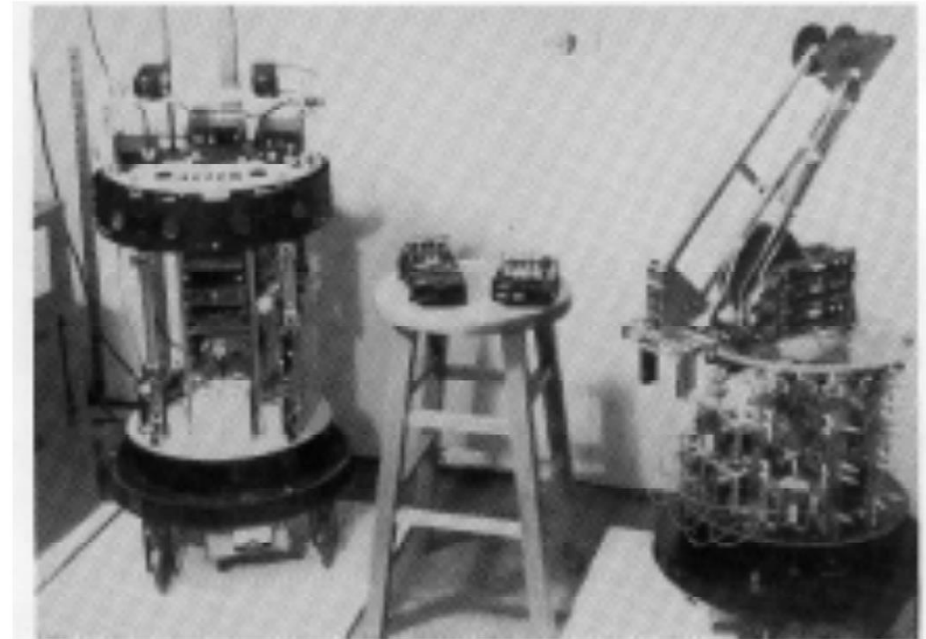


Fig. 1. The four MIT AI laboratory Mobots. Left-most is the first built Allen, which relies on an offboard LISP machine for computation support. The right-most one is Herbert, shown with a 24 node CMOS parallel processor surrounding its girth. New sensors and fast early vision processors are still to be built and installed. In the middle are Tom and Jerry, based on a commercial toy chassis, with single PALs (Programmable Array of Logic) as their controllers.

Inside a Brooks abstraction

- Each layer is a fixed-topology network of small asynchronous finite-state machines.
- Machines communicate on fixed wires using short messages: 1 bit on small robots, 24 bits on larger ones.
- There is no global data store, central control, or dynamically created communication link.
- Suppression redirects an input; inhibition blocks an output for a specified duration.

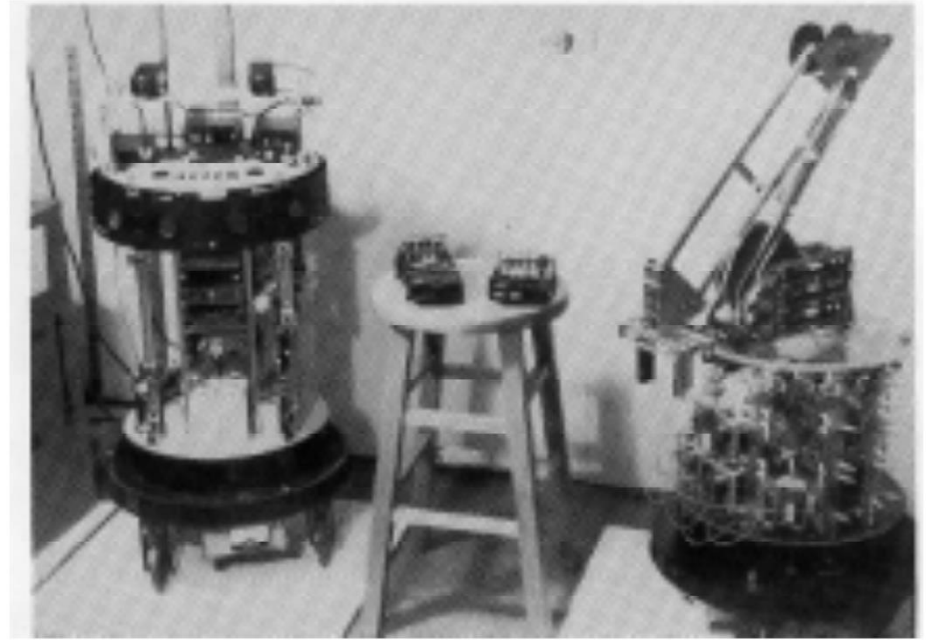


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Elephants don't play chess

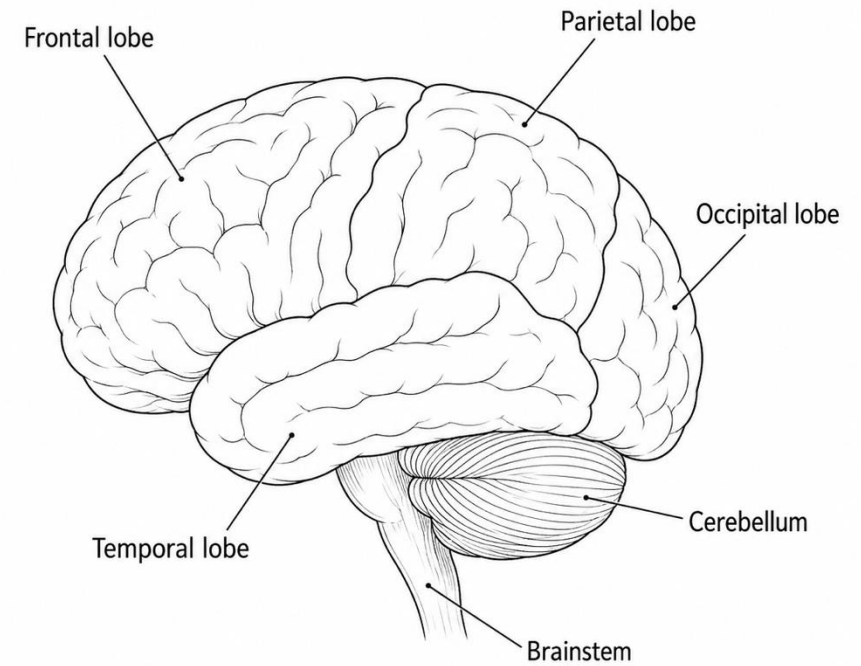
Intelligence developed in mobile, situated organisms; abstraction came later and should not be the starting assumption.

Sensors and actuators first

- Typed symbolic input/output is rejected as ungrounded; goals must become physical action.
- ‘The world is its own best model’: sense it appropriately and often enough.
- Build complete agents bottom-up and evaluate them in dynamic, uncontrolled environments.
- Open limits include scaling many behavior modules, multisensory fusion, learning, and automatic interface construction.

Discussion

Discussion: *Do we need to understand something before we can simulate it?*



Discussion: *Do we need to understand something before we can simulate it?*



Functionalism
vs.
Biological Naturalism